CS & IT ENGINEERING

Programming in C

Control Flow Statements

Lec- 04 Switch Statements

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TOPICS TO BE COVERED

- Switch-Case Statements

switch statement

* Keyword : used to create selection statements with multiple choice.

* Multiple choices are provided by another keyword 'case'.

```
1
switch (n) {

    case 1 :   Code we want to execute if the value of
               n is 1
               break;

    case 2 :   Code we wont to execute if the value of
               n is 2
               break;

    default :  Code we wont to execute if n does not
               match with any case label
               break;
}
```

2+3

switch(expression/condition){

Evaluated

case $constant_1$: Code1

break;

5

case $constant_2$: Code2

break;

case $constant_3$: Code3

break

default : Code4

break;

}

Three

Step 1

```
int i = 3;            3
switch(i){
    case 2 :          printf("Two");
                      break;
    case 3 :          printf("Three")
                      break;
    default :         printf("wrong");
                      break;
}
```

```
int i=3;          3
switch(i){

case 3 :          printf("Three");

case 2 :          printf("Two");

default :         printf("wrong");

}
```

Three Two wrong

$\text{switch(expression)} \rightarrow \text{int}$

$\rightarrow$ switch(2+3) $\rightarrow$ ✓

switch (2) $\rightarrow$ ✓

switch (~2) $\rightarrow$ ✓

int i=2, j=3;

$\text{switch}(\underbrace{i+j\times 2}_{\text{int}})$ ✓

$\text{switch}(\underbrace{\underline{12.3}_{\text{float}} + \underline{2}_{\text{int}}}_{\text{float}}) \rightarrow$ Invalid

switch(printf("Pankaj")) $\rightarrow$ ✓

printf("%d", 'A'); valid 'A' → 65

```
switch('A')
{

}
```

✓

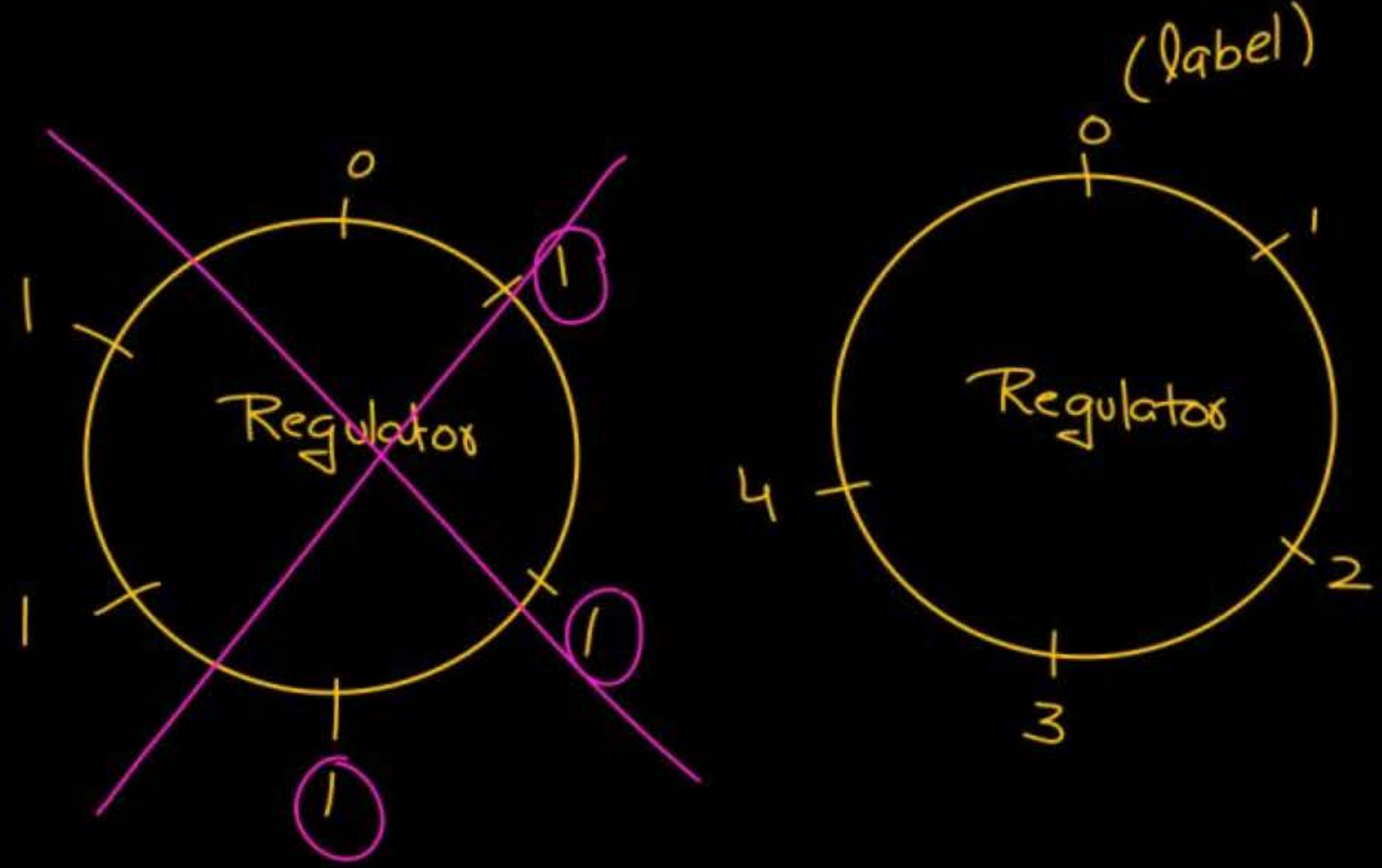
```
switch('A'+2)
{

}
```

mandatory

```
switch( )
{

}
```



① No duplicate case labels are allowed

```
switch( !2 + 5){

    case 0 :  printf("Hello");
              break;

    case 1 :  printf("One");
              break;
                 1
    Case !2||2.3 :  printf("OR")
                    break;

    default :  printf("Wrong");
               break;
}
```

Error

duplicate case labels

```
int i=2 ;
switch(i){

    case 3 :
        break;

    case 1 :
        break;

    case 2 :
        break;

    default :
        break;
}
```

```
switch(i){

    default :
        break;

    case 3 :
        break;

    case 1 :
        break;

    case 2 :
        break;
}
```

```
int i = 3;        3
  switch(i){
    default : printf("0");
              break;

    case 2 :  printf("2");
              break;

    case 1 :  printf("1");
              break;
  }
```

Position of default does not matter.

```c
if(1)
    printf("1");
else if(2)
    printf("2");
else if(3)
    printf("3");
```

```c
int i=3;
switch(i){
    case 2 : printf("2");
             break;
    case 1 : printf("1");
             break;
}
```

default is optional

```
int i=5;
switch(i){

}
```

Valid

Dummy

```
switch(i);
```

```
switch( );
```

Mandatory

Error

```
int i=2,j=4;
switch(i){

    case j :
            break;

    case 3 :
            break;

    case 2+3 :
            break;

}
```

must be literals

```
int i = 3;

switch(i){
        i = i+2 ;
      case 3 : printf("3");
               break;
      case 5 : printf("5");
               break;
      }
```

ignore this

```
int i=3;
switch(i){

    case 1 : printf("1");
             break;
             i = i+2;        ignored

    case 5 : printf("5");
             break;

    case 3 : printf("3");          3
             break;

}
```

if (i>=10 && i<=20)
{

}
else if (i >= 30 && i<=100)
{

}

range

only on few compiler

switch(i){

1space

case 10 ... 20 :

low

high

break;

$-\sim 11 \Rightarrow - (\sim 11)$

$-(11+1)$

$-(-12)$

$\Rightarrow 12$ ✓

$6-3 << 2$

$3 << 2 \Rightarrow 3 \times 2 \times 2 \Rightarrow 12$

```
switch(i) {   // 1

   default :
        break;

   case 1 :
        break;

   Case 2 :
        break;
}
```

1	
2	
other	

```
for(i=1; i<=n; i=i×3)
   {
    for(j=i; j<=n; j++)
        {
        printf("Pankaj");
        }
   }
```

$i=1$	$i=3^1$	$i=3^2$	$\cdots$	$i=3^k$
1 to n	3 to n	3^2 to n		3^k to n
$(n-1+1)$	$(n-3+1)$	$n-3^2+1$		$(n-3^k+1)$

k+1 ⇒ parenthesis

$$(n-1+1)+(n-3+1)+(n-3^2+1)+\cdots+(n-3^k+1)$$

$$(n+1)(k+1)-1-3^1-3^2\cdots-3^k$$

$$(n+1)(k+1)-(1+3^1+3^2+\cdots+3^k)$$

$$(n+1)(k+1)-\frac{3^{k+1}-1}{3-1}$$

$$=(n+1)(1+\lfloor\log_3 n\rfloor)-\left(\frac{3^{1+\lfloor\log_3 n\rfloor}-1}{2}\right)$$

$3^k<=n$

$k<=\log_3 n$

$k=\lfloor\log_3 n\rfloor$

$$(n+1+x)+(n+1+2x)+(n+1+3x)$$

$$3(n+1)+(x+2x+3x)$$

$$(n+1-1)+(n+1-3^1)+(n+1-3^2)+(n+1-3^3)+\dots(n+1-3^k)$$

$$(k+1)(n+1)-\underbrace{(1+3^1+3^2+\dots+3^k)}$$

case 100 . . . 20 X

```
if ( i==12 || i==15 || i==18 )
    {
        code
    }
else if ( i==10 || i==14 || i==17 )
    {

    }
```

12, 15, 18

10, 14, 17

```
{1, 3, 9} →
{2, 4, 8} →

switch(i) {

    case 1 :

    case 3 :

    case 9 :  printf("Pankaj");
              break;

    case 2 :

    case 4 :

    case 8 :  printf("sharma");
              break;
}
```


THANK YOU GW
SOLDIERS !